

Introduction to Computing (ITC)

What is Computer Science?

Science of Computing ?

What is Computer Science?

For other uses, see [Computer science \(disambiguation\)](#).

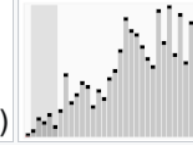
Computer science is the study of [computation](#), [information](#), and [automation](#).^{[1][2][3]} Computer science spans [theoretical disciplines](#) (such as [algorithms](#), [theory of computation](#), and [information theory](#)) to [applied disciplines](#) (including the design and implementation of [hardware](#) and [software](#)).^{[4][5][6]}

[Algorithms](#) and [data structures](#) are central to computer science.^[7] The [theory of computation](#) concerns abstract [models of computation](#) and general classes of [problems](#) that can be solved using them. The fields of [cryptography](#) and [computer security](#) involve studying the means for secure communication and preventing [security vulnerabilities](#). [Computer graphics](#) and [computational geometry](#) address the generation of

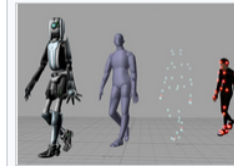
Fundamental areas of computer science

$0 := \lambda f. \lambda x. x$
 $1 := \lambda f. \lambda x. f \ x$
 $2 := \lambda f. \lambda x. f \ (f \ x)$

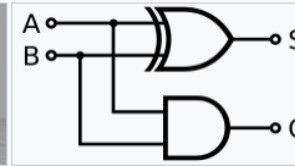
[Programming language theory](#)



[Computational complexity theory](#)



[Artificial intelligence](#)



[Computer architecture](#)

Computer science

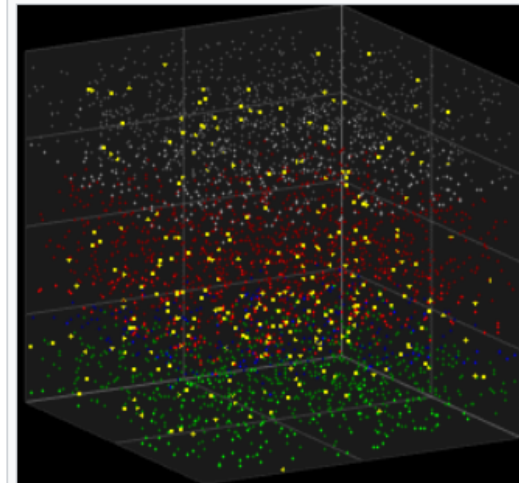


https://en.wikipedia.org/wiki/Computer_science

What is Computing?

Computing is any goal-oriented activity requiring, benefiting from, or creating [computing machinery](#).^[1] It includes the study and experimentation of [algorithmic](#) processes, and the development of both [hardware](#) and software. Computing has scientific, engineering, mathematical, technological, and social aspects. Major computing disciplines include [computer engineering](#), [computer science](#), [cybersecurity](#), [data science](#), [information systems](#), [information technology](#), and [software engineering](#).^[2]

The term *computing* is also [synonymous](#) with [counting](#) and [calculating](#). In earlier times, it was used in reference to the action performed by [mechanical computing machines](#), and before that, to [human computers](#).^[3]



Computer simulation, one of the main cross-computing methodologies 

<https://en.wikipedia.org/wiki/Computing>

What is Computing?

- It is derived from the Latin terms - **com (together) and putare (to settle)**.
- To compute means given one or more inputs, to settle at an output (which is exactly what a computer or any computing device does).

In other words, to compute means **to reason**. It consists of broadly two kinds of reasoning:

numerical reasoning

analytical/logical reasoning.

- **A computing device enables us to outsource our mental work to it the way physical devices like automobiles, wheels, pulleys etc., enable us to outsource our physical work to them**

History of Computing

4000-1200 B.C. Inhabitants of the first known civilization in Sumer keep records of commercial transactions on clay tablets.

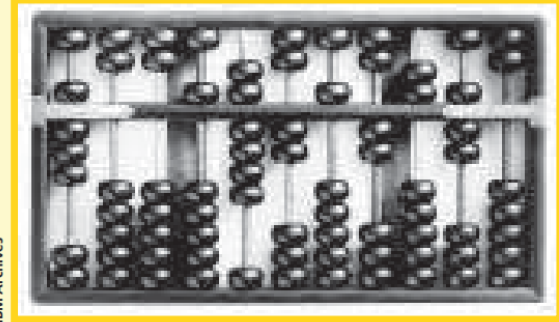


The University Museum, University of Pennsylvania

3000 B.C. The abacus is invented in Babylonia.

250-230 B.C. The Sieve of Eratosthenes is used to determine prime numbers.

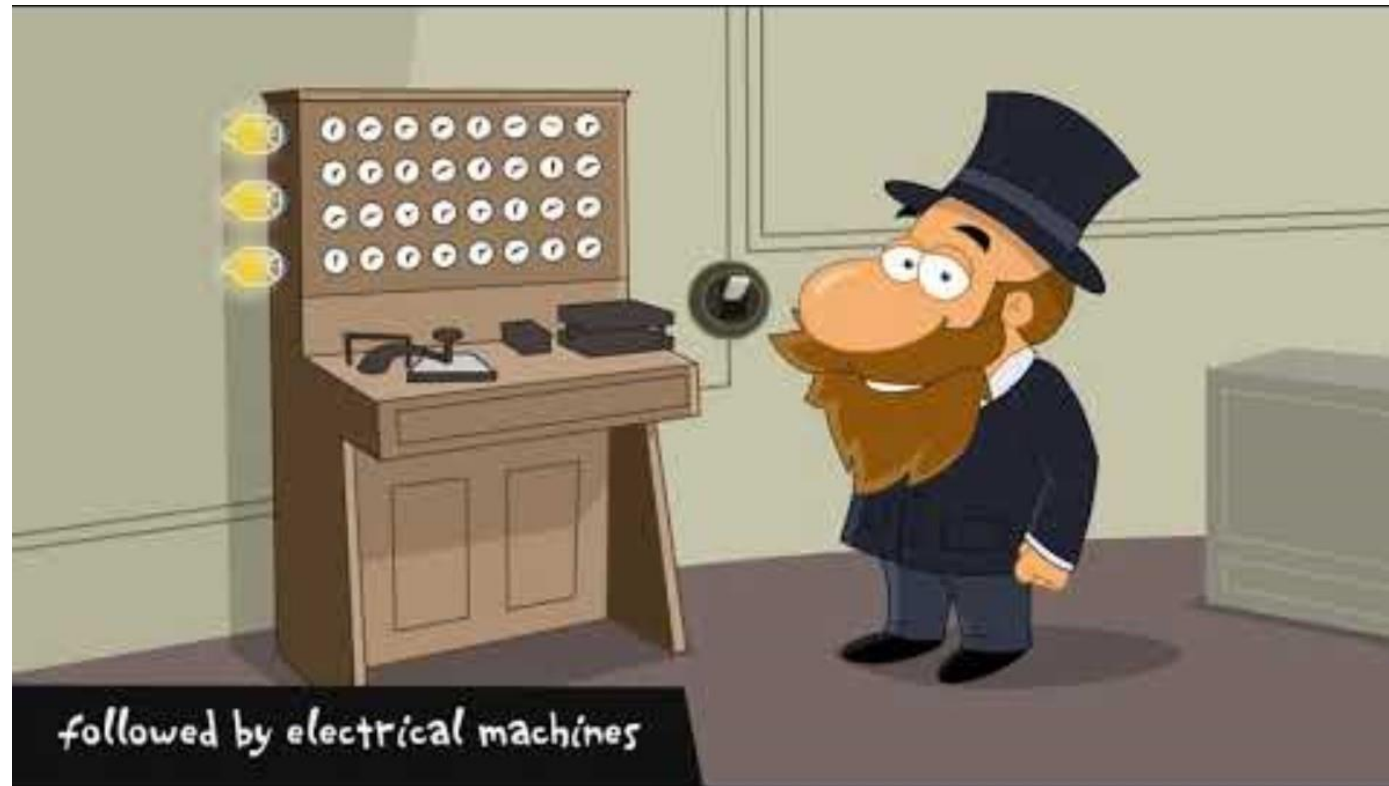
About 79 A.D. The "Antikythera Device," when set correctly according to latitude and day of the week, gives alternating 29- and 30-day lunar months.



IBM Archives

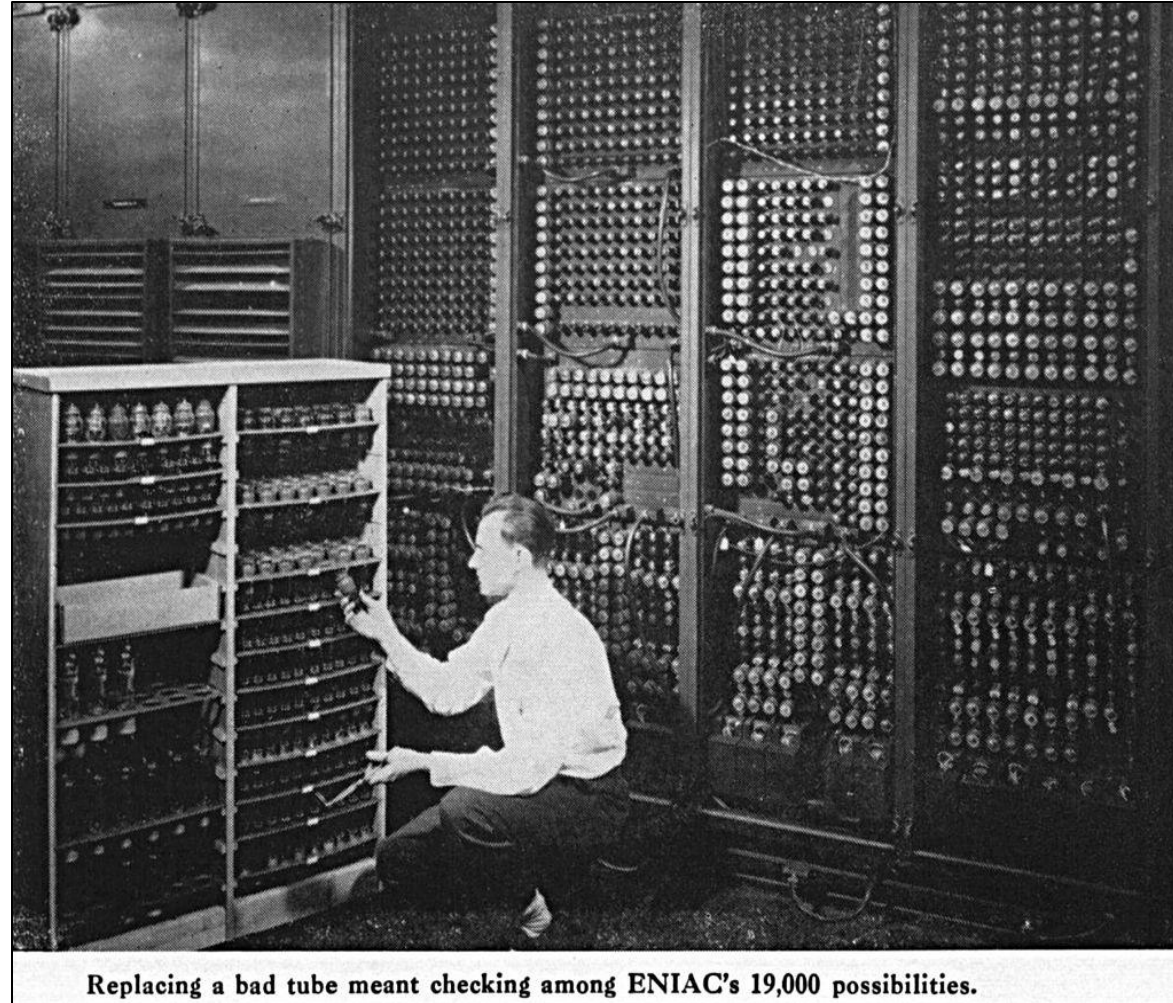
About 1300 The more familiar wire-and-bead abacus replaces the Chinese calculating rods.

History of Computing (in nutshell)

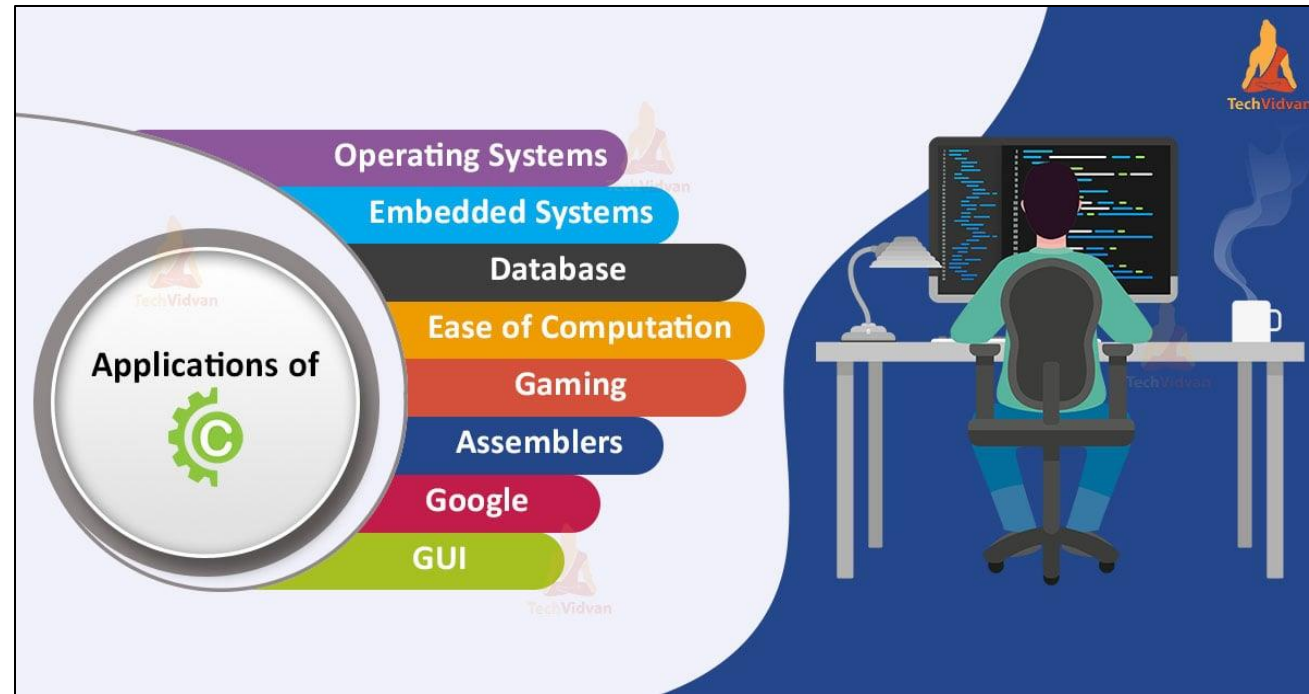


Video source: <https://youtu.be/EDpwP3fEJ24>

Problem with ENIAC



Applications



Topics covered till now

- Introduction to course 
- Course Syllabus
- Course Logistics
- What is Computing
- History of Computing
- Applications

Next topic.....

- Binary digits
- Number system
 - Base Conversions

Binary digit

The only language that a computer understand is actually
Binary dig**its**, i.e., Bits

Binary Digit



A 10x10 grid of binary digits (0s and 1s) displayed in green on a black background. The digits are arranged in a regular pattern, with each row containing 10 digits and each column containing 10 digits. The digits are 00111100011100101111110001110, 00011111001111111101111110000, 11110111101111111111100011111, 01110110000001001100111011111, 10000011101111101110111110111, 10001001001111100010001100001, 11001100101110011111111111111, 11110000100001010111111110001, 11000010011110010000110000000.

Binary Digit



Binary numbers

- Binary numbers are the flow of information in the form of zeros and ones used by digital computers and systems

Binary + Digit = Bit

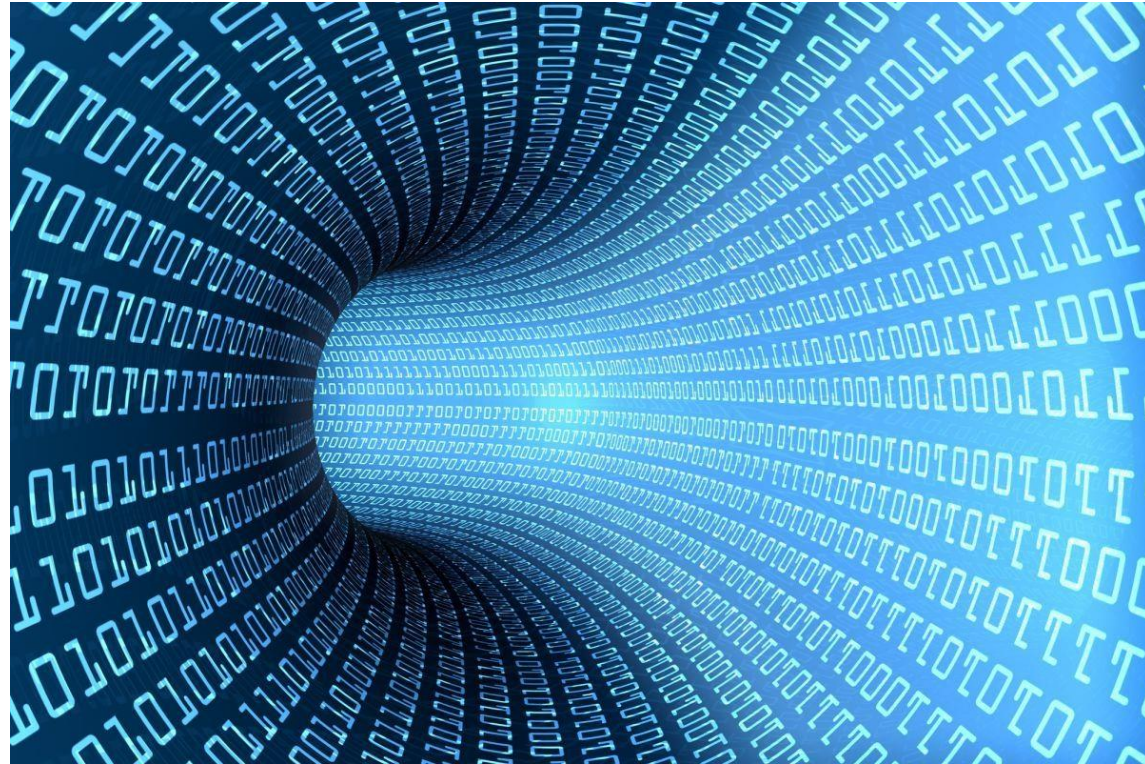
What is so 'binary' about a computer?



Why do we keep seeing pictures like these?

Binary + Digit = Bit

What is so 'binary' about a computer?



and this?

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?**

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?** The condition is that you cannot speak!

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?** The condition is that you cannot speak!

- Write the msg on a piece of paper.

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?** The condition is that you cannot speak!

- Write the msg on a piece of paper.
- Fold the paper

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?** The condition is that you cannot speak!

- Write the msg on a piece of paper.
- Fold the paper make a paper rocket!

Sending some text across..

Suppose you have a piece of message that you want to send across to someone sitting at the other corner of the room **What can you do?** The condition is that you cannot speak!

- Write the msg on a piece of paper.
- Fold the paper make a paper rocket! and the throw!

Sending some text across..

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- Pretty simple, no?!

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Now, suppose the text is as long as harry potter -

Sending some text across..

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Now, suppose the text is as long as harry potter - all volumes included!

- Throw the books at the person?

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- Fold the paper make a paper rocket! and the throw!
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Now, suppose the text is as long as harry potter - all volumes included!

- Throw the books at the person?
- How about a less savage idea?

An idea!

An idea!

① If we can..

An idea!

- 1 If we can.. convert the entire text into bits (0's and 1's).

An idea!

- ① If we can.. convert the entire text into bits (0's and 1's).
- ② Raise your hand for 1 and keep it down for 0.

An idea!

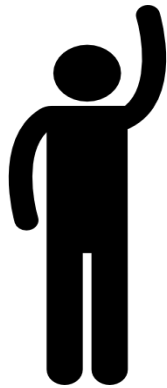
- ① If we can.. convert the entire text into bits (0's and 1's).
- ② Raise your hand for 1 and keep it down for 0.
- ③ Tell your friend you will stay in each pose for 2 seconds.

An idea!

- If we can.. convert the entire text into bits (0's and 1's).
- Raise your hand for 1 and keep it down for 0.
- Tell your friend you will stay in each pose for 2 seconds.
- And Shoot!

An idea!

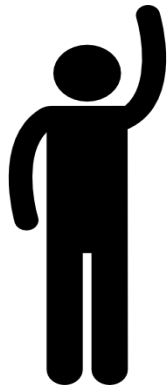
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1

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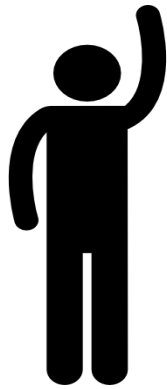
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0

An idea!

- ① If we can.. convert the entire text into bits (0's and 1's).
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1



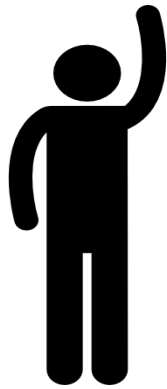
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0

An idea!

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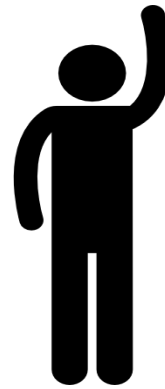
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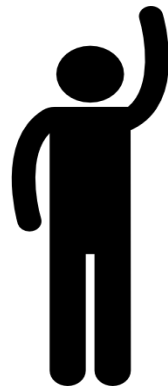
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1

An idea!

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- ② Raise your hand for 1 and keep it down for 0.
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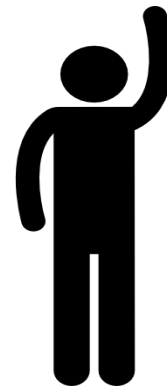
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0



0



1

...

Idea works .. but

Idea works .. but

Two problems..

Idea works .. but

Two problems..

- 1 How can we convert text to binary bits?

Idea works .. but

Two problems..

- ① How can we convert text to binary bits?
- ② How many bits do all volumes of Harry Potter amount to?
Will your hand survive?

Simple Encoding of text

Convert these numbers into binary

A	1
B	2
C	3
D	4
E	5
F	6
G	7
H	8
I	9
J	10
K	11
L	12
M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

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E	5
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G	7
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M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

a	27
b	28
c	29
d	30
e	31
f	32
g	33
h	34
i	35
j	36
k	37
l	38
m	39
n	40
o	41
p	42
q	43
r	44
s	45
t	46
u	47
v	48
w	49
x	50
y	51
z	52

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J	10
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L	12
M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

a	27
b	28
c	29
d	30
e	31
f	32
g	33
h	34
i	35
j	36
k	37
l	38
m	39
n	40
o	41
p	42
q	43
r	44
s	45
t	46
u	47
v	48
w	49
x	50
y	51
z	52

!	53
"	54
#	55
\$	56
%	57
&	58
'	59
(	60
)	61
*	62
+	63
,	64
-	65
.	66
/	67
:	68
;	69
<	70
=	71
>	72
?	73
@	74
\	75

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A	1
B	2
C	3
D	4
E	5
F	6
G	7
H	8
I	9
J	10
K	11
L	12
M	13
N	14
O	15
P	16
Q	17
R	18
S	19
T	20
U	21
V	22
W	23
X	24
Y	25
Z	26

a	27
b	28
c	29
d	30
e	31
f	32
g	33
h	34
i	35
j	36
k	37
l	38
m	39
n	40
o	41
p	42
q	43
r	44
s	45
t	46
u	47
v	48
w	49
x	50
y	51
z	52

!	53
"	54
#	55
\$	56
%	57
&	58
'	59
(	60
)	61
*	62
+	63
,	64
-	65
.	66
/	67
:	68
;	69
<	70
=	71
>	72
?	73
@	74
\	75

SPACE	76
NEWLINE	76
START	78
STOP	79

Digitization of text.



1



0

001000|011011|101100|101100|110011 → Harry

Digitization of text - ASCII

- ASCII (American Standard Code for Information Interchange, pronounced “ASKee”) requires seven bits for each character.

ASCII Table

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	\$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

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M	13
N	14
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r	44
s	45
t	46
u	47
v	48
w	49
x	50
y	51
z	52

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"	54
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\$	56
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'	59
(	60
)	61
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Digitization of text.



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001000|011011|101100|101100|110011 → Harry

Digitization of text - ASCII

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3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	\$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
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11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

Digitization of text - Extended ASCII

- Extended ASCII can accommodate extra characters and requires 8 bits for each character.

[illegible]

Only text?

‘Anything’ can be digitized, so long as it is information!

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Multimedia:

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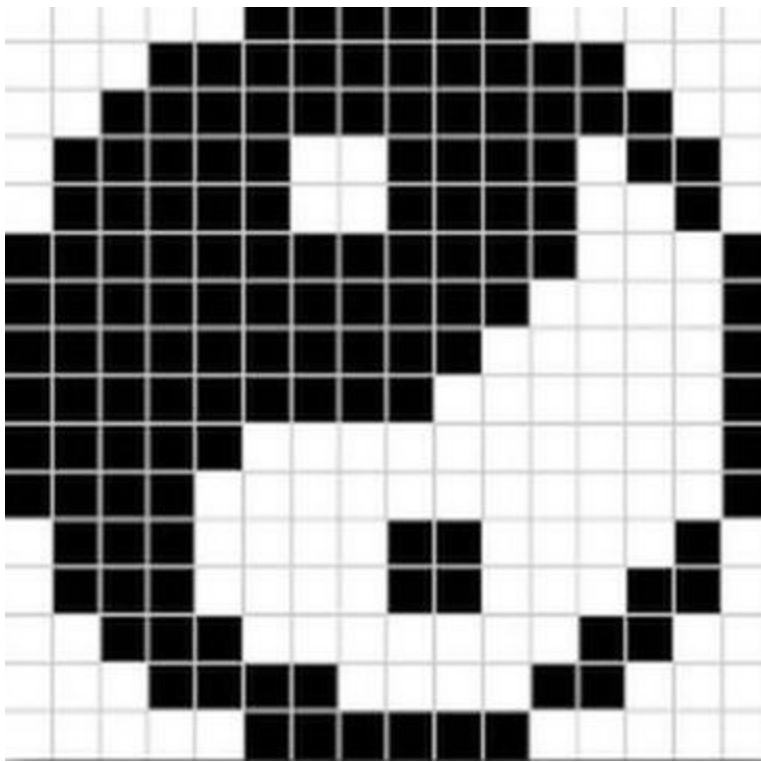
- **Multimedia:**

- Pictures
- Music
- Videos

- **Serious Stuff:**

- Documents
- Numbers
- Emails
- Bank transactions
- Webpages etc.

Some binary images!



16 × 16 pixels

Some binary images!



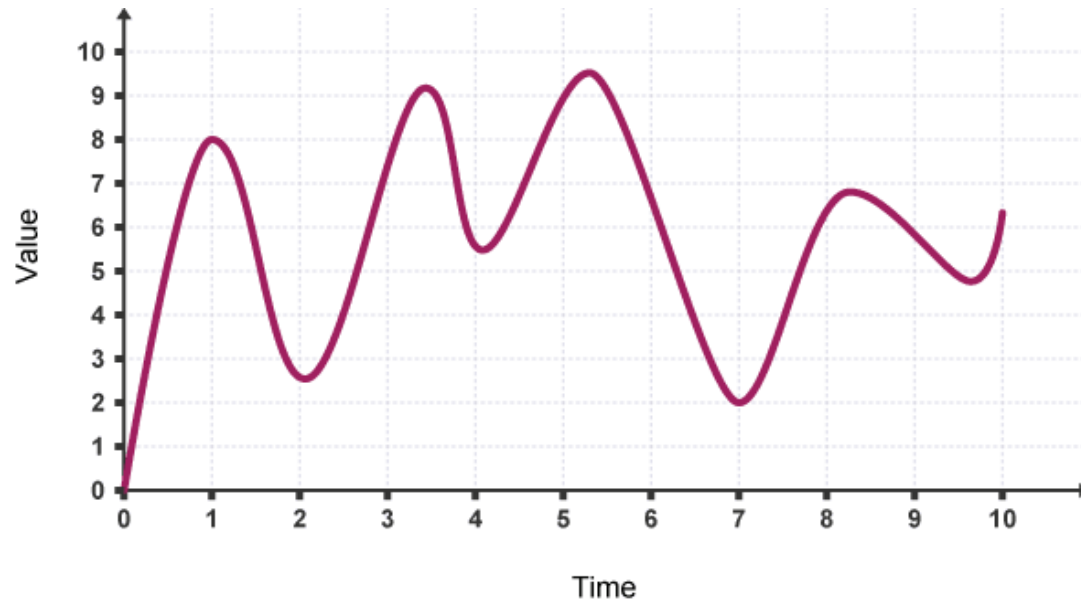
More pixels

Some binary images!

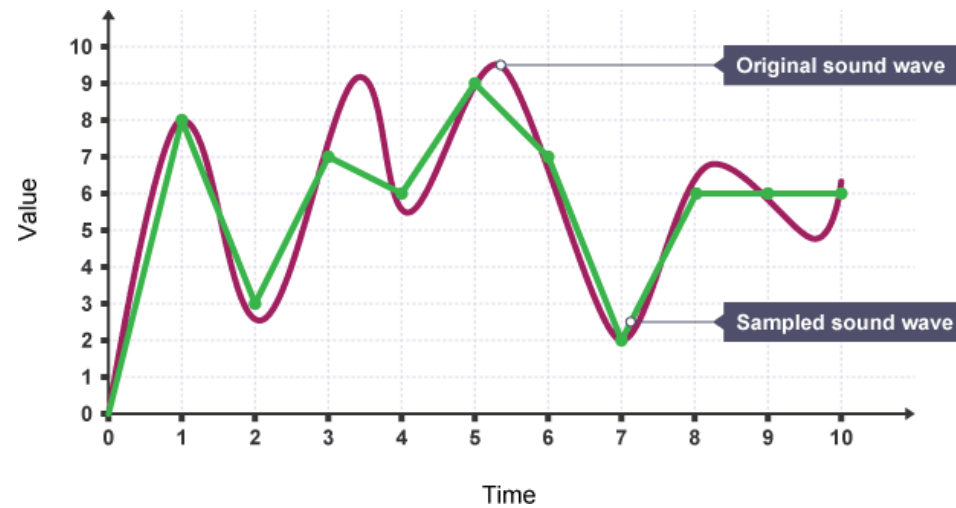
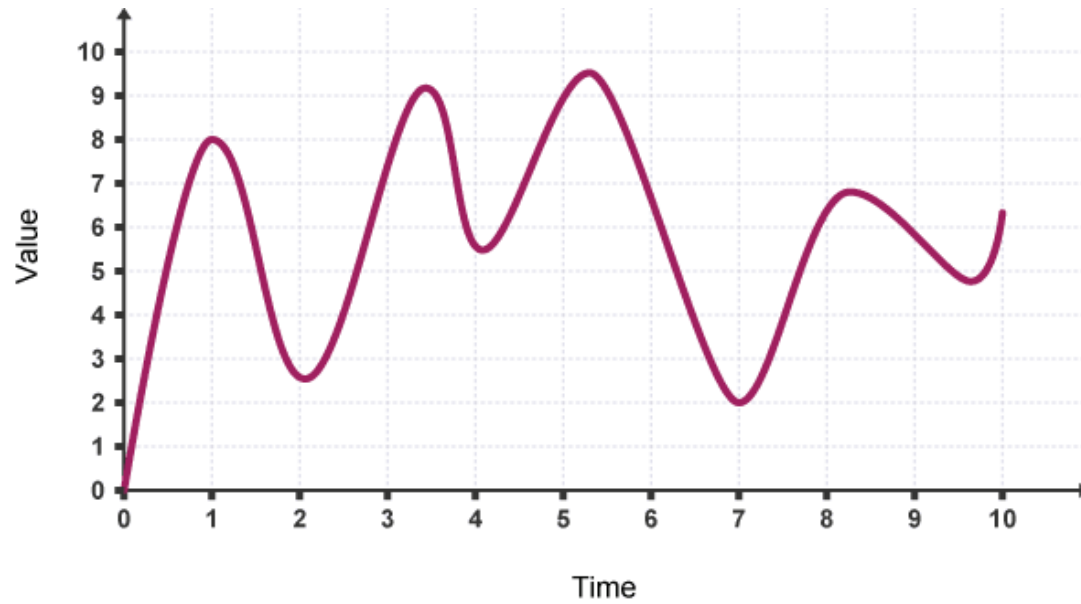


roughly 500×500 pixels
00000001010101110101...

How about audio?

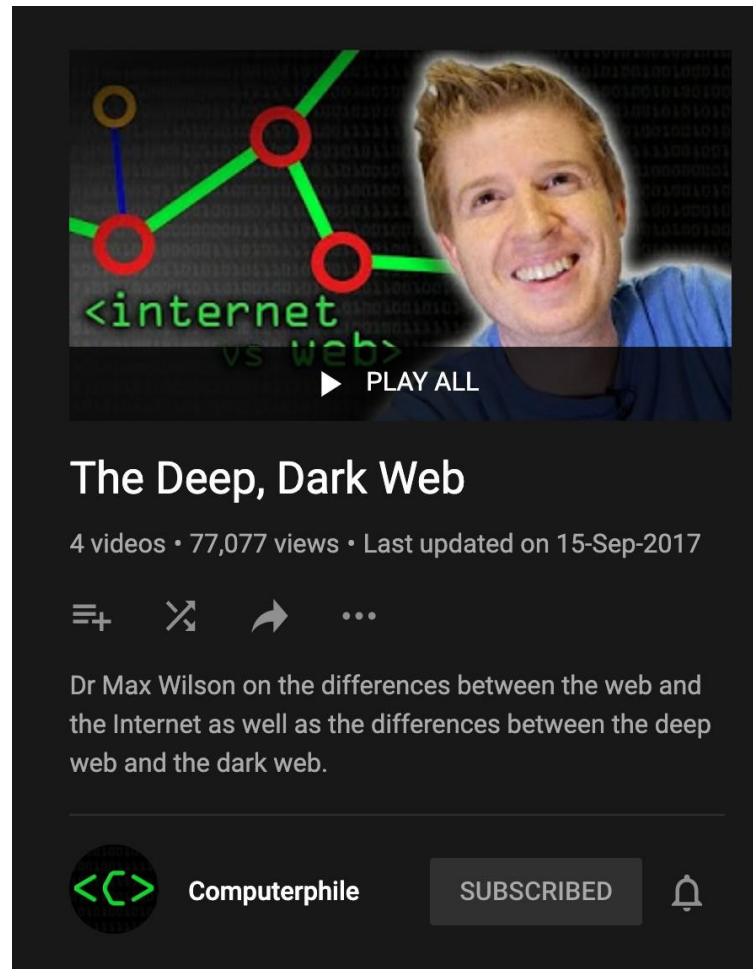


How about audio?



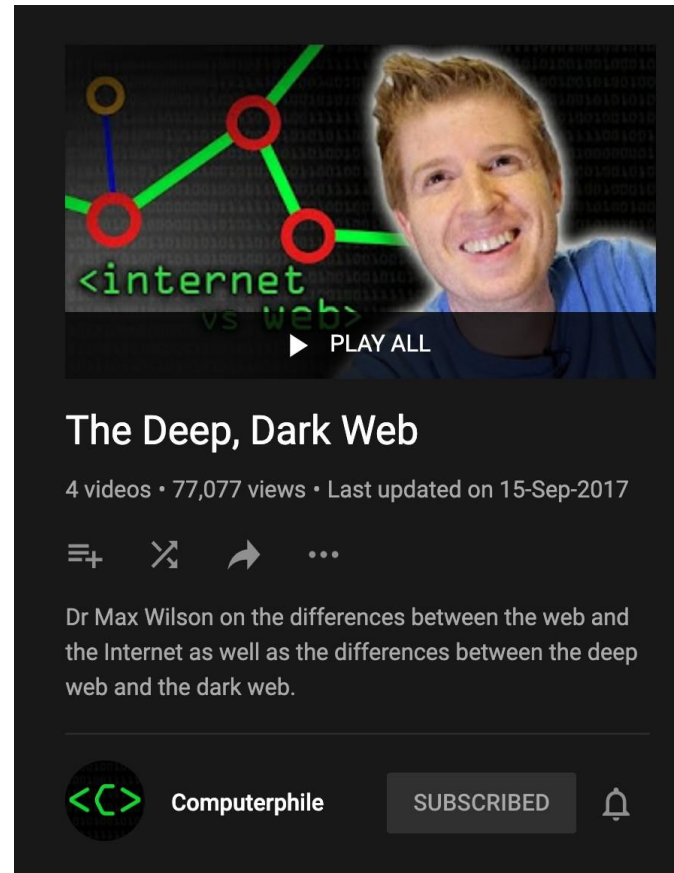
8,3,7,6,9,7,2,6,6,6 = 1000,0011,0111...

and also videos..



Computerphile is an awesome youtube channel.

and also videos..



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A video is but a series of images (**frames**)

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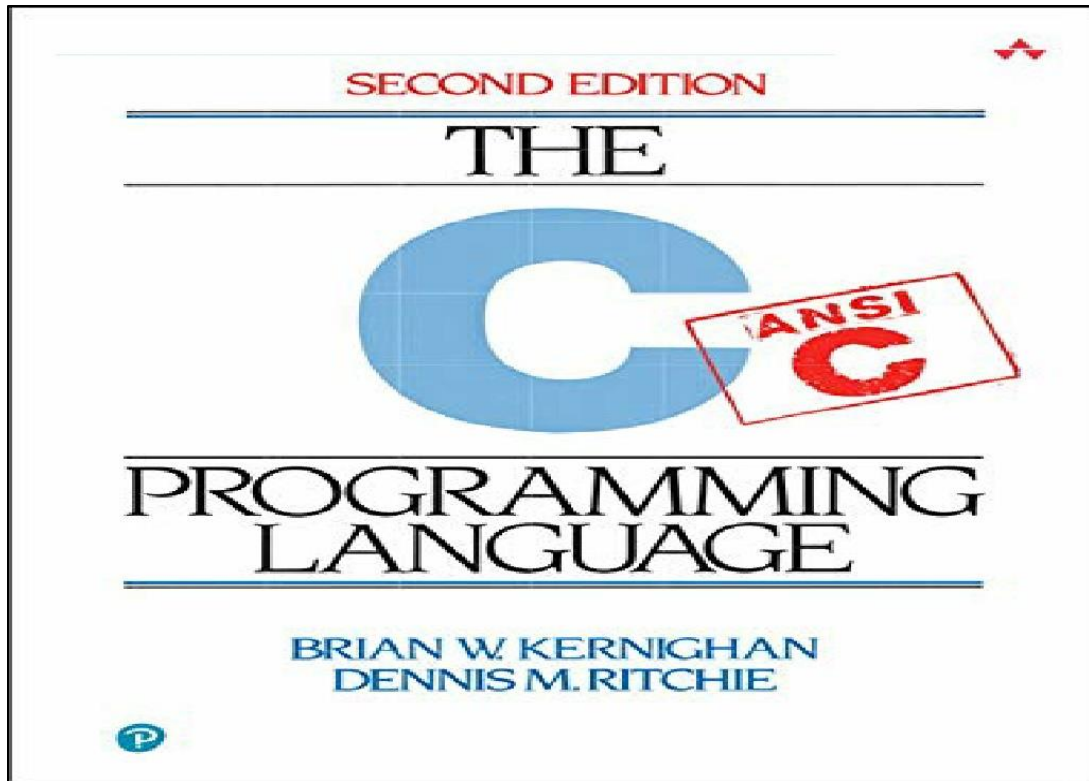
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- ⑥ More on that in the next class.

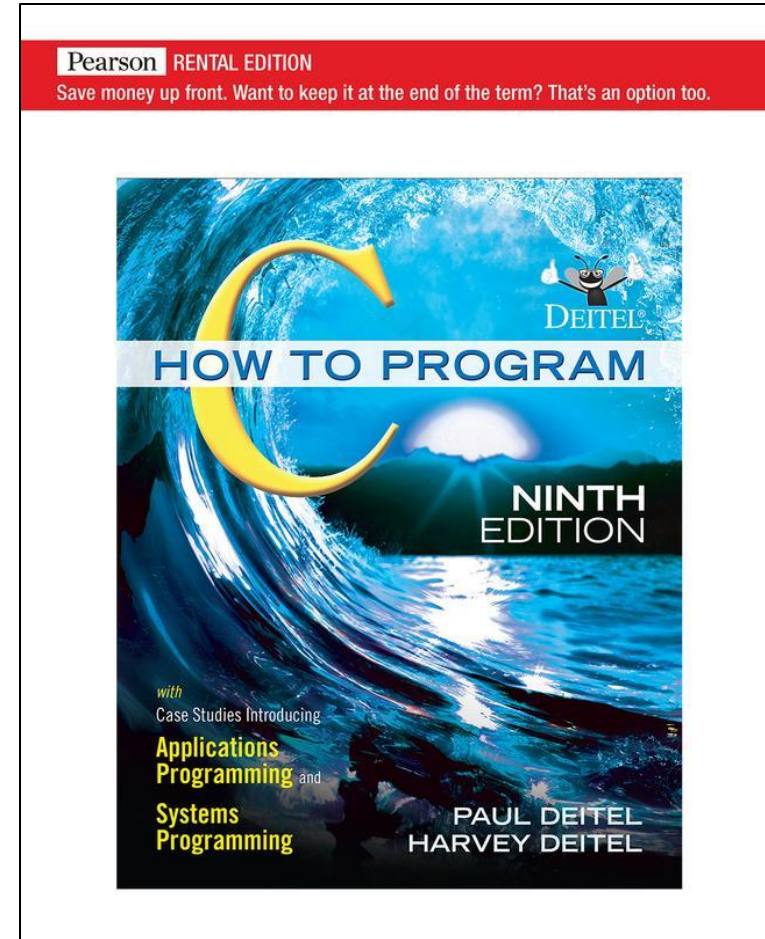
Acknowledgements

- Lecture slides modified from resources by Dr. Rajesh and Dr. Bhanu from Mahindra University.

Reference books/ materials



“The C Programming Language” by Brian Kernighan and Dennis Ritchie



“C How to Program” by Paul Deitel and Harvey Deitel